Akilan Jayakumar Deepa

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Executive Summary

AI Master's student at Brunel University London with a strong background in computer science and applied AI. Skilled in Python, OpenCV, and LLMs. Experienced in developing intelligent, user-focused systems in chatbot, media processing, and civic automation domains. Experienced in writing clean, maintainable, production-oriented code through real-world projects. Passionate about building efficient, accessible, and scalable AI-driven applications.

Skills

Languages: C/C++, Python, JavaScript, SQL
Technologies & Tools: LM Studio, TensorFlow, Scikit-learn, OpenCV, MoviePy, ReactJS, Node.js, HTML5, CSS3, Webflow, GitHub, VS Code, Figma, MySQL, Firebase, Microsoft Excel, PowerPoint, Word, Object-Oriented Programming (OOP).

Project Works

AI-Powered Customizable Chatbot

- Led Development of an offline chatbot using a pre-trained LLM integrated with LM Studio.
- Added a GPU utilization control feature, enabling users to select GPU load for faster response generation. Enabled multi-domain conversations and chat log downloads for reusability.
- Delivered scalable AI assistance offline with optimized GPU performance and user control.
- Python, LM Studio

Media Processing & Real-Time Editing System

- Developed a media editor supporting live image adjustments and video operations (merge, trim, resolution).
- Led the project including architecture, logic, and UI implementation. Delivered optimized real-time performance using Python libraries. Streamlined media editing workflows, reducing manual editing time.
- Python, OpenCV, MoviePy.

Public Issue Reporting System (Ongoing)

- Designed a full-stack web app for citizens to report local issues with geo-tagging. Developed modules for complaint submission, officer assignment, and live tracking.
- Planning AI integration for auto-categorization, urgency detection, and heatmap analytics. Simplified public reporting and improved case tracking efficiency.
- React, Node.js, MySQL/Firebase

Education

Brunel University London, United Kingdom Sep 2025 - Present
MSc Artificial Intelligence
Relevant Coursework: Advanced Artificial Intelligence, Machine Learning Applications, Deep Learning and Neural Networks, Modern Data, Quantitative Data Analysis (QDA), Artificial Intelligence.

SRM Institute of Science and Technology, Chennai, India Aug 2021 - May 2025
B.Tech Computer Science and Engineering (Specialization in AI & ML) ***CGPA: 8.80 / 10***
Relevant Coursework: Object Oriented Programming, Databases, Discrete Mathematics, Data Structures and Algorithms, Artificial Intelligence and Machine Learning, Database Management Systems (DBMS), Web Application Development, Data Analytics and Visualization, Human–Computer Interaction (HCI).

Awards and Certificates

- Completed *Generative AI with Large Language Models* course by DeepLearning.AI on Coursera, gaining hands-on experience in LLMs, prompt engineering, and applied machine learning using PyTorch.
- Completed *Foundations of User Experience (UX) Design* by Google on Coursera, focusing on user-centric design, accessibility, and design sprint methodology.